

ANESS RAHMANI

Machine Learning Engineer | Computer Vision | Full-Stack

d.rahmanianess@gmail.com • (+213) 669 23 13 86 • github.com/anness793 • [LinkedIn](#) • anness-rahmani.site

ML engineer with hands-on experience building end-to-end computer vision systems, from data collection and model training to full-stack deployment. Built and defended AcciSense, a real-time road accident detection and severity classification platform using Video Transformers, achieving 87%+ accident detection accuracy on a custom dataset. Fluent in Python, PyTorch, and modern web technologies (FastAPI, React). Trilingual: Arabic (native), English (C2), French (B2-C1).

TECHNICAL SKILLS

ML / AI: PyTorch, Deep Learning, Computer Vision, Video Transformers (SlowFast R50, Video Swin), 3D-CNN (R3D-18), YOLO, Model Quantization & Pruning

Backend: Python, FastAPI, Node.js, Express.js, RESTful APIs, MQTT (Mosquitto), WebSocket

Frontend: React, Vite, Angular, JavaScript, TypeScript, HTML, CSS

Databases: MongoDB, SQL, Supabase

Tools: Git, MATLAB, Docker (basic), CLI, Kaggle GPU training (2x NVIDIA T4)

Languages: Arabic (native), English (C2), French (B2-C1)

PROJECTS

AcciSense — Road Accident Detection Platform | *Python · PyTorch · FastAPI · React · MQTT · Supabase* | 2025 – 2026

- Designed and built an end-to-end real-time accident detection system comparing 3 deep learning architectures (R3D-18, SlowFast R50, Video Swin Transformer) on a custom 4-class severity dataset (BTVAD)
- Achieved 87%+ accuracy on binary accident detection; best severity model (R3D-18) reached 63.55% accuracy and 63.10% macro F1 on a challenging 4-class imbalanced dataset
- Engineered full production platform: FastAPI backend, React/Vite frontend, MQTT autonomous monitoring via Mosquitto, WebSocket real-time updates, Supabase logging
- Implemented automated PDF incident reports (triggered on major detections), ntfy.sh push notifications, and model quantization/pruning for edge deployment optimization
- Trained on Kaggle using 2x NVIDIA T4 GPUs; evaluated with confusion matrices, per-class metrics, ROC curves, and Cohen's Kappa

Custom YOLO Object Detection Pipeline | *Python · YOLO · Computer Vision* | 2024

- Built end-to-end custom object detection pipeline: data annotation, YOLO model training on custom dataset, accuracy evaluation and testing

IoT Device Monitoring Dashboard | *FastAPI · React · MQTT · WebSocket* | 2023

- Built real-time monitoring system for IoT devices with FastAPI backend, React frontend, and MQTT subscriber for continuous device data streaming and live visualization

EXPERIENCE

Network Engineering Intern | [Algérie Télécom](#) | Dec 2025 – Mar 2026 | Khemis Miliana, Algeria

- Worked hands-on with transmission infrastructure and end-to-end telecom network systems
- Gained operational knowledge of transmission protocols used in national telecom operations

Data Annotation & Labeling Specialist | [Freelance \(Upwork / LinkedIn\)](#) | 2022 – 2024 | Remote

- Completed multiple data annotation, proofreading, and audio recording contracts for AI training datasets via Upwork and direct clients

- Cleaned and standardized a database of thousands of marine species records for a research project, resolving inconsistent classification schema

IT Department Head | [University Club](#) | 2021 – 2023 | Khemis Miliana

- Led technical initiatives and managed digital infrastructure for university organization

EDUCATION

Master's Degree in Telecommunication Systems | [Université Djilali Bounaama](#) | 2024 – 2026 | Algeria

- Thesis: Automatic Road Accident Detection and Severity Estimation Using Video Transformers
- Relevant coursework: Machine Learning, 5G Networks, Advanced Signal Processing, Network Security

Bachelor's Degree in Electrical Engineering | [Université Djilali Bounaama](#) | 2020 – 2023 | Algeria

- Graduation project: Battery Energy Storage System with Capacitor for Renewable Energy Integration (MATLAB simulation)